

First Terminal Examination – 2080

Class : XI

Subject: Physics (1011)

Full Marks: 75

Time: 3 hrs.

SET 'B'

Pass Marks: 30

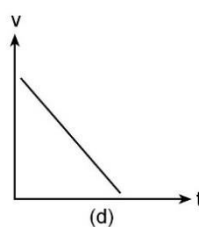
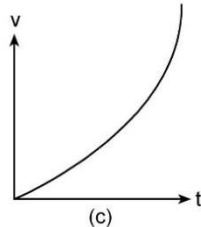
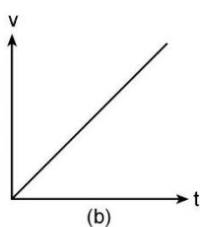
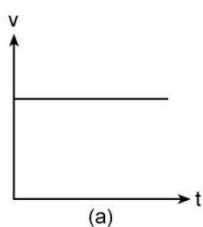
Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group A

Rewrite the best alternative of each question in your answer sheet. [11 × 1 = 11]

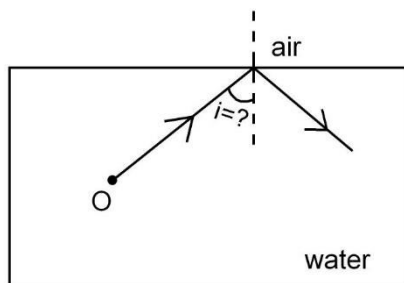
1. The dimensions $[ML^{-1}T^{-2}]$ may correspond to
 - a. Work
 - b. linear momentum
 - c. pressure
 - d. energy per unit area
2. A meter ruler is used to measure the length of a piece of string in a certain experiment. It is found to be 20 cm long to the nearest millimeter. How should this result be recorded in a table of results?
 - a. 0.2000 m
 - b. 0.200 m
 - c. 0.20 m
 - d. 0.2 m
3. If two vectors each having magnitude 'P' is acting at a point and angle between them is θ , then the resultant is given by
 - a. $\sqrt{2} P \cos \frac{\theta}{2}$
 - b. $2 P \cos \frac{\theta}{2}$
 - c. $\sqrt{2} P \sin \frac{\theta}{2}$
 - d. $2 P \sin \frac{\theta}{2}$
4. Which of the following velocity-time graph represents the zero



acceleration?

5. The number of significant digits in the measurement 15.00 mm are
a. 2 b. 3 c. 4 d. 5
6. A steel-tape gives correct measurement at 20°C . A piece of wood is being measured with the steel-tape at 0°C . The reading is 25 cm on the tape. The real length of the given piece of wood must be:
a. 25 cm b. less than 25 cm
c. more than 25 cm d. can not say
7. The focal length of a convex mirror is maximum for
a. Red colour light b. Yellow colour light
c. Violet colour light d. same for all colour light
8. Light of wavelength $6000^{\circ}\text{\AA}$ in air enter a medium of refractive index 1.5. the frequency of light in the medium is
a. $6000 \times 10^{10} \text{ Hz}$ b. $3 \times 10^8 \text{ Hz}$
c. $5 \times 10^{14} \text{ Hz}$ d. none

9. From the figure, what condition of



- a. $i = c$ b. $i > c$ c. $i < c$ d. none of above
10. Which of the statement is always true?

- a. Real depth = Apparent depth b. Real depth < Apparent depth
 c. Real depth > Apparent depth d. Real depth = thickness of medium

11. A diver in water at a depth 1 m sees the whole outside world through horizontal circle of radius when the refractive index is μ .

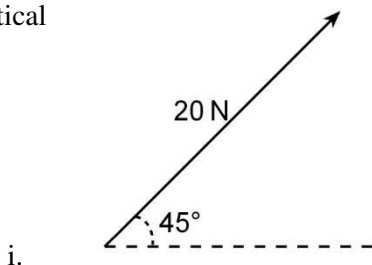
- a) Fehler! b) Fehler! c) Fehler! d)
 Fehler!

Group – B

Give short answer to the following questions.

[8 × 5 = 40]

12. a. Calculate the horizontal and vertical components of the following. [2]



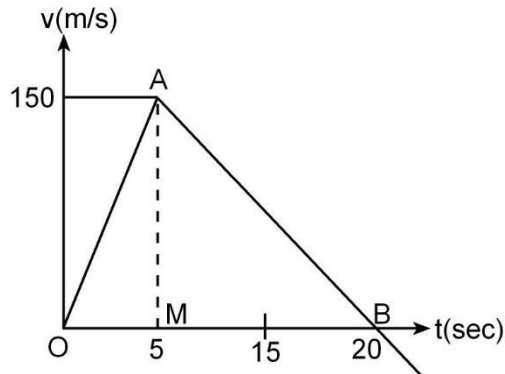
- b. A rocket fires two engines simultaneously. One produces a thrust of 725N directly forward, while the other gives a 513N thrust at 32.4° above the forward direction. Find the magnitude and direction of the resultant force that these engines exert on the rocket. [3]

13. a. State triangle law of vector addition. Using this law, find the magnitude

and direction of the resultant of two vectors $\vec{A}$ and $\vec{B}$ inclined at an angle θ from each other. [1+2]

- b. A force of $(7\hat{i} + 6\hat{k})$ Newton makes a body move on a rough plane with a velocity of $(3\hat{j} + 4\hat{k}) \text{ ms}^{-1}$. Estimate the power. [2]

14. A rocket is fired vertically about 5 sec. It attains a velocity 150 m/s.



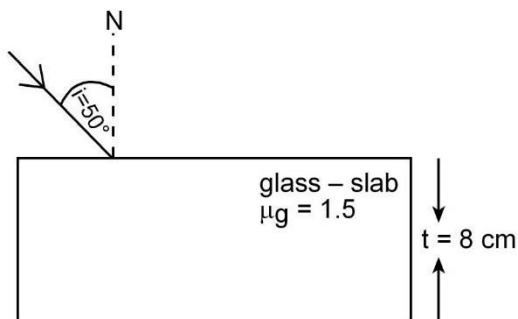
- What is the acceleration of the rocket during OA ? [1]
 - What is the physical significance of the line AB ? [1]
 - Find the change in velocity with respect to time during AB . [2]
 - If area of triangle OAB gives maximum height reached by rocket, estimate the value of maximum height. [1]
15. a. A scientific book describes a new temperature scale called Z , in which boiling and freezing points of water are referred as $65^\circ Z$ and $-15^\circ Z$ respectively. To what temperature on Fahrenheit scale would a temperature -9
- Two beakers of water, A and B, initially are at the same temperature. The temperature of the water in beaker A is increased by $10^\circ F$ and the temperature of the water in beaker B is increased by $10K$. After these temperature changes,
 - Which beaker of water has the higher temperature? Explain. [2]
 - What about their heat energy? [1]

16. The first pendulum clock was invented in 1656 A.D. by Dutch scientist and inventor Christiaan Huygens, and patented the following year. Huygens was inspired by investigations of pendulums by Galileo Galilei beginning around 1602 A.D. Galileo discovered the key property that makes pendulums useful timekeepers.



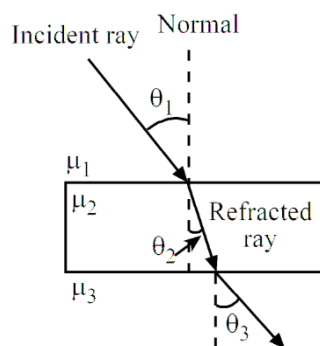
- a. Why is invar used in making a clock pendulum? [1]
 - b. Pendulum clocks generally run fast in winter and slow in summer, why?
 - c. We can use a bimetallic strip made of brass and steel each of different length whose length would change in such a way that difference between their lengths remain 10 cm for all temperature. What should be length of each strip ? [Given, linear expansivity of brass = $18 \times 10^{-6} / ^\circ\text{C}$ and Steel = $12 \times 10^{-6} / ^\circ\text{C}$] [3]
17. a. What is mirror formula? Derive it for concave mirror when virtual image is formed. [1+2]
- b. A convex mirror of radius of curvature 20 cm forms an image which is half the size of object. Locate the position of the object and image. [2]

18. From the given figure



- a. Complete the figure with ray diagram when the glass slab produce the lateral shift (d) [1]
- b. Define lateral shift. [1]
- c. Derive the relation for lateral shift produced by the glass slab of thickness (t) [2]
- d. Find the angle of emergent. [1]

19. a. Why do in summer, roads often appear to be covered with water when seen from a distance? Explain. [2]
- b. A light ray of wavelength 589 nm (Produced by a sodium lamp) traveling through air is incident on a smooth, flat slab of crown glass ($\mu = 1.52$) at an angle θ_1 of 30.0° to the normal, as sketched in figure.
 - i. Find the angle of refraction, θ_2 .
 - ii. At what angle θ_3 does the ray leave the glass as it re-enters the air?
 - iii. How does the answer for θ_3 change if the ray enters water ($\mu = 1.33$) below the slab instead of the air?



Group – C

Give long answer to the following questions.

[3 × 8 = 24]

20. a. Use velocity- time graph to obtain equation of motion.
 - i.e $v^2 = u^2 + 2as$ [3]
- b. A man can swim at the rate of 6 km hr^{-1} in still water. A river 1 km wide flows at the rate of 3 km hr^{-1} .
 - i. Swimmer wishes to cross the river straight, along what direction must he strike? [2]
 - ii. How much time he would take to cross the river? [1]

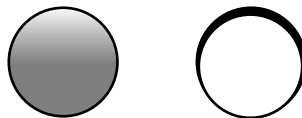
- c. Check the dimensional correctness of the equation, $t = 2\pi \sqrt{\frac{m}{k}}$ **Fehler!** where 't' be the time period, 'm' is the mass and 'k' is the force per unit length. [2]

21. a. A metal cube is heated from $\theta_1^\circ \text{C}$ to $\theta_2^\circ \text{C}$ so that length increases from l_1 to l_2 and volume from V_1 to V_2 as
 $l_2 = l_1 [1 + \alpha(\theta_2 - \theta_1)]$
 and $V_2 = V_1 [1 + \gamma(\theta_2 - \theta_1)]$

i. Define the α and γ used in above expressions. [2]

ii. Show that $\alpha = \frac{\gamma}{3}$ **Fehler!** [2]

- b. Two bodies made of the same material have the same external dimensions and appearance, but one is solid and the other is hollow as shown in figure. When they are heated to the same temperature range, is overall volume expansion the same or different? explain [1]



- c. The volume of a metal sphere is increased by 1% of its original volume, when it is heated from 20°C to 220°C . Calculate the coefficient of linear, superficial and cubical expansion of the metal. [3]

22. a. For a prism, prove that,
 $\mu = \frac{\sin i}{\sin r}$ **Fehler!** [3]
- b. Define critical angle and total internal reflection with conditions for occurrence of total internal reflection. [1+2]
- c. A transparent cube of 12cm edge contains a small air bubble. Its apparent depth when viewed through one face of the cube is 6cm and when viewed through the opposite face is 2cm. What is the actual distance of the bubble from the first face? [μ for glass = 1.5] [2]

ॐ - The End - ॐ